AuraRoute AI

An advisory AI agent that makes cross-border tuition payments safe to plan

AI as a Financial Participant | Topic C: Cross-border & Student Finance Assistant

2026 Shenzhen FinTechathon, International Track | Team AuraRoute AI

THE PROBLEM

Paying tuition across borders hides three risks

1. Rules

An annual foreign-exchange quota of USD 50,000 per person applies. Above it, a bank needs supporting documents. Splitting payments to stay under limits is something SAFE has acted against.

2. Cash

Paying the invoice can leave a student below a living-expense or proof-of-funds buffer. Few tools check the balance after paying.

3. Time

Cross-border transfers take days to clear. A late payment can trigger a registration hold, so students need a safe latest start date.

Students today piece this together from bank pages, forums and guesswork.

Sources: SAFE Q&A on the annual quota (2007); SAFE circular on fund splitting (2009); Bank of China guidance on tuition documents.

AI AS A FINANCIAL PARTICIPANT

AI proposes, code decides, the student approves

The model reads

When an invoice layout breaks our parser, a model extracts the fields. It sees redacted text only, has no tools, and its output is schema-checked. It can only fill gaps.

Code decides

Currency conversion, quota check, illustrative balance baseline and latest safe start date are computed by deterministic code, with each rule shown with its source.

The student approves

Every parsed value is confirmed at a validation gate. Nothing moves downstream, and the orchestrator refuses unverified data.

Read > Redact > Confirm > Check > Advise | Advisory and read-only: never moves money

SYSTEM DESIGN

Architecture: data and rule lookups travel apart

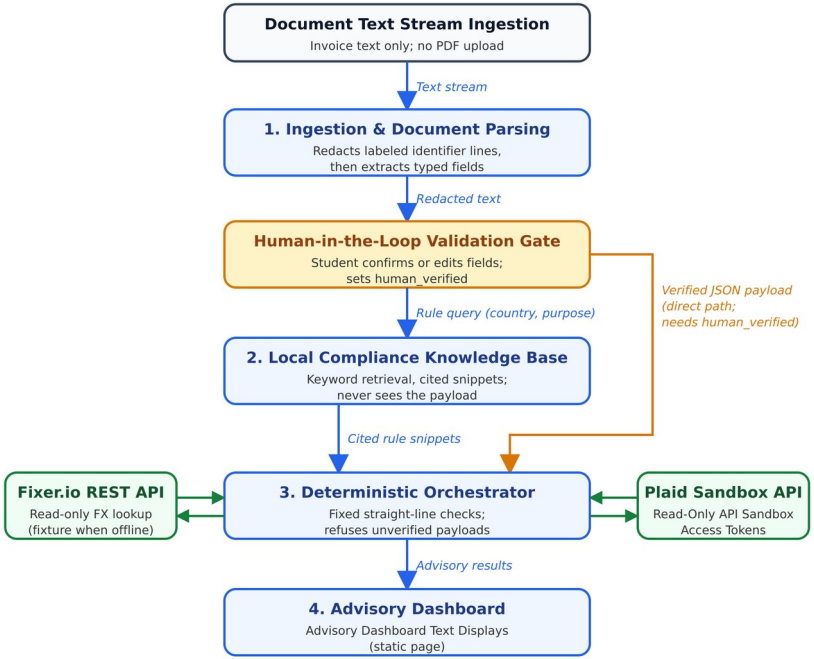


Figure 1. System architecture. The orange path carries the verified payload directly to Layer 3.

Parallel routing

Verified payload goes straight to the orchestrator; only a rule query reaches the rule base.

Read-only tools

Exchange-rate lookup and sandbox balance read. No payment or write endpoint exists.

Tamper-evident audit

Hash-chained log of every check and every blocked payload.

TASK COMPLETION FRAMEWORK

Two problems caught before any money moves

15,240 USD

Invoice of 12,000 GBP, converted by the read-only FX tool

Over quota

Exceeds the 8,000 USD left. Advises the documented bank route. Never splits.

1,260 USD

Balance after paying, below the 2,500 USD illustrative framework baseline. Warning raised.

10 Dec

Latest safe start for a 15 Dec due date, with a 5-day clearing buffer

Evidence we submit

- 29 unit tests pass, including blocked unverified payloads and hostile model output
- End-to-end run log from the actual code, with a verified audit chain
- Fixer.io and Plaid Sandbox supported with keys; recorded fixtures offline

SECURITY & COMPLIANCE INFRASTRUCTURE

Security and compliance are built into the flow

No money movement

No payment or write endpoint. Only an FX read and a balance read.

Redaction first

Names, passport, phone, email, government ID, address, student ID and birth date removed before parsing or any model call.

Untrusted model output

Strict schema and allow-list. Cannot overwrite parsed values or verify itself.

Human gate

No analysis until the student confirms. The orchestrator re-checks.

No structuring

Over the quota, we point to the documented bank route and never suggest splitting.

Tamper-evident audit

Hash-chained entries. Alteration of earlier entries is detected.

Permission Tiers and a 14-item Known-Risk List (Item 5). Prompt injection is reduced, not eliminated.

INNOVATION & INTERACTION

What is new, what is built, what is next

What is new

A guardrail, not an optimizer: it prevents structuring instead of enabling it.

A bounded AI role: the model reads, code decides.

Honesty by design: sources, residual risks and data modes are shown.

Built and tested

Redaction, human gate, rule retrieval with citations, FX and balance tools, quota and baseline checks, advisory dashboard, hash-chained audit log, optional Gemini or Anthropic extraction fallback.

Roadmap

Verified rule corpus with article citations.

Plaid Link, authentication and encryption at rest.

More countries and embedding-based retrieval.

Pilot with a university international office.

AuraRoute AI: plan the payment before you make it.